

Extension of Scribble-Supervised LiDAR Semantic Segmentation

Semester Project Proposal
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September 29, 2022

GROUP MEMBERS

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I. DESCRIPTION OF THE PROJECT

Scribble-supervised LiDAR semantic segmentation behaved well on specific scribble dataset. However, labeling points manually is very expensive. Then, we want to develop an automatic point labeler to generate high quality scribble labels. After that, we will reproduce the scribble-supervised pipeline on new scribble datasets to see if this pipeline is generally applicable.

II. WORK PACKAGES AND TIMELINE

A. The goal of the project

- 1) Develop a point labeler which can generate scribble labels automatically using full labels.
- 2) Improved the quality of the scribble labels generated by point labeler.
- 3) Reproduce scribble-supervised pipeline on new scribble datasets generated by proposed point labeler.

B. Timeline

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| Stage 1: Preparation for Semester Project | 28.09.2022 - 23.10.2022 (4 weeks) |
| <ul style="list-style-type: none"> • Set up Ubuntu environment, learn 3D RANSAC and point labeler. • Learn how to scribble large-scale real data by point labeler. • Read relevant literature about 2D and 3D point labeler. • Improve the algorithm for scribble small objects, such as cars, trunks. | |
| Stage 2: Main phase of Semester Project | 24.10.2022 - 29.01.2023 (14 weeks) |
| <ul style="list-style-type: none"> • Develop the algorithm for automatic point labeler, especially for roads and facades of buildings. • Evaluate and improve the quality of the scribble labels generated by this point labeler. • Reproduce scribble-supervised pipeline on new scribble datasets. | |

III. OUTCOMES AND DEMONSTRATION

A. Outcomes

- 1) A fine point labeler that can generate scribble labels automatically.
- 2) A new scribble dataset generated by point labeler.

B. Demonstration

- 1) Demonstration of the process of point labeling on large-scale new dataset.
- 2) Related paper about this project.